

Graded potentials

The written notes for the graded potentials walkthrough. Week 4 competencies 12 and 14.

To print: these notes as a two-column PDF. Already printed the Week 4 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

Two kinds of electrical signal

	Graded potential	Action potential
Where	Input zone: dendrites and cell body	Starts at the trigger zone, travels the axon
Size	Varies with the strength of the stimulus	Always the same size (all or none)
Distance	Short; gets smaller with distance	Long; does not get smaller
Threshold	None	Generated only if the trigger zone reaches threshold
Channels	Chemically (ligand) gated or mechanically gated	Voltage-gated sodium and potassium

Graded potentials on a dendrite

1. A stimulus opens gated channels: a neurotransmitter binds to chemically gated channels at a synapse, or stretch opens mechanically gated channels in a sensory ending.
2. Ions cross, and the membrane potential moves toward the equilibrium potential of the ion whose permeability increased.
3. Sodium entering makes the inside less negative: a depolarizing graded potential. At a synapse this is an excitatory postsynaptic potential (EPSP), because it brings the neuron closer to firing an action potential.
4. Potassium leaving makes the inside more negative than rest: a hyperpolarizing graded potential. At a synapse this is an inhibitory postsynaptic potential (IPSP), because it moves the membrane potential farther from threshold and makes the neuron less likely to fire.
5. The channels close and the membrane returns to rest. A graded potential lasts from a few milliseconds to tens of milliseconds.
 - A stronger stimulus opens more channels or holds them open longer, so more ions cross and the change in voltage is larger. That is what graded means: the amplitude varies with the strength of the stimulus.
 - Example amplitudes recorded at one site: weak stimulus 3 mV, medium 7 mV, strong 12 mV; a stimulus that opens potassium channels, about 8 mV below rest.
 - In a sensory receptor, a graded potential produced by a stimulus such as stretch is called a receptor potential.

Graded potential

A small change in membrane potential confined to one region of the membrane, whose amplitude varies with the strength of the stimulus.

EPSP (excitatory postsynaptic potential)

A depolarizing graded potential produced by a neurotransmitter at a synapse.

IPSP (inhibitory postsynaptic potential)

A hyperpolarizing graded potential produced by a neurotransmitter at a synapse.

Receptor potential

A graded potential produced in a sensory receptor by a stimulus such as stretch.

Why graded potentials fade

- Local current flow: the positive charge that enters spreads along the inside of the dendrite in both directions, away from where it entered.
- Current leak: some of that charge leaves across the membrane through the open leak channels along the dendrite, the same leak channels that set the resting potential.
- Cytoplasmic resistance: the cytoplasm, which contains many proteins and other molecules, resists the flow of charge through it.
- Because of current leak and cytoplasmic resistance, the graded potential is smaller at every point farther from where it started. This loss of amplitude with distance is decremental conduction. Example: 20 mV at point 1 next to the open channels, then 12, 7 and 4 mV at points farther along.
- A wider dendrite has a larger cross section for charge to flow through, so its cytoplasmic resistance is much lower. Current leak does not decrease; a wider dendrite has a little more membrane, and so more leak channels, along each length. Cytoplasmic resistance falls much more than leak rises, so the graded potential spreads farther.

Local current flow

Movement of charge through the cytoplasm away from where ions entered.

Current leak

Loss of charge across the membrane through open leak channels.

Cytoplasmic resistance

The cytoplasm's resistance to the flow of charge; lower in a wider process.

Decremental conduction

The decrease in a graded potential's amplitude with distance from where it started.

What reaches the trigger zone

- The trigger zone is the axon hillock and initial segment. An action potential is generated there only if the membrane potential at the trigger zone reaches threshold, about -55 mV. From rest at -70 mV that takes about 15 mV of depolarization.
- The outcome depends on the amplitude of the graded potential when it arrives at the trigger zone, not its amplitude where it started.
- Graded potentials that overlap in time add together: summation. EPSPs move the membrane potential toward threshold and IPSPs move it away. An action potential is generated only if the net depolarization reaches threshold.
- Spatial summation: graded potentials from different synapses that are active at the same time add together.
- Temporal summation: one synapse is activated again before its previous graded potential has faded, and the two add together.
- Real graded potentials that overlap add up to a little less than their simple sum, but the reasoning is the same.

Inputs reaching the trigger zone	Arithmetic	Voltage	Result
One EPSP, 8 mV (20 mV at the synapse)	$-70 + 8$	-62 mV	Subthreshold, no action potential
Two EPSPs at the same time, 8 mV each	$-70 + 8 + 8$	-54 mV	Suprathreshold, an action potential is generated
Two EPSPs of 8 mV plus one IPSP of 5 mV	$-70 + 8 + 8 - 5$	-59 mV	Subthreshold, no action potential

Threshold

The membrane potential at the trigger zone, about -55 mV, at which an action potential is generated.

Subthreshold

Too small to bring the trigger zone to threshold.

Suprathreshold

Large enough to bring the trigger zone to threshold.

Summation

The adding together of graded potentials that overlap in time.

Spatial summation

Summation of graded potentials from different synapses active at the same time.

Temporal summation

Summation of graded potentials from one synapse activated again before its previous graded potential has faded.

Patient: the pufferfish meal

- Tetrodotoxin blocks voltage-gated sodium channels from the outside of the cell. It does not block chemically gated or mechanically gated channels.
- His resting potential is normal (-70 mV). A weak stimulus still produces a small graded potential, about 5 mV, that fades.

- A strong stimulus produces about 20 mV, which goes past threshold, but no action potential appears, because the action potential depends on the voltage-gated sodium channels that the toxin blocks.
- His neurons can still produce graded potentials in response to input, but they cannot generate action potentials to carry signals along their axons. The action potential walkthrough finishes this case.